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MILITARY SPECIFICATION

CABLE, SPECIAL PURPOSE, ELECTRICAL WM-46/U

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of ten conductors, Buna S insulated and jacketed cable designated as Cable, Special Purpose, Electrical WM-46/U. Eight of the conductors are equivalent to No. 20 AWG, four of which are shielded, and two of the conductors are equivalent to No. 16 AWG, one of which is shielded.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

J-C-98	— Cable and Wire, Insulated, Methods of Sampling and Testing.
DDD-T-86	— Tape; Cotton, Bleached, Dyed, or Grey, General Use.
PPP-T-60	— Tape; Pressure Sensitive Adhesive, Waterproof — For Packaging and Sealing.
PPP-B-585	— Boxes; Wood, Wirebound.
PPP-B-591	— Boxes; Fiberboard, Wood-Cleated.
PPP-B-601	— Boxes, Wood, Cleated Plywood.
PPP-B-621	— Boxes; Wood, Nailed and Lock-Corner.
PP-B-636	— Boxes; Fiber.
PPP-B-640	— Boxes; Folding, Fiberboard, Corrugated, Triple Wall.

MILITARY

MIL-P-116	— Preservation, Methods of.
MIL-I-631	— Insulation, Electrical, Synthetic - Resin Composition, Nonrigid.

FSC 6145

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MIL-W-3861 — Wire, Electrical (Bare Copper).

MIL-I-3930 — Insulating and Jacketing Compounds, Electrical (For Cable, Cord, and Wire).

MIL-R-10409 — Reels, Cable (Steel).

MIL-R-12083 — Reels, Cable (Wood).

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL MOTOR FREIGHT CLASSIFICATION NUMBERS A-3 AND 14.

(Application for copies of the above publication should be addressed to the American Trucking Association, Inc., Traffic Department, 1424 16th Street, N. W., Washington 6, D. C.)

UNIFORM FREIGHT CLASSIFICATION NUMBER 14.

(Application for copies should be addressed to Uniform Freight Classification Committee, 202 Union Station, Chicago 6, Ill).

CONSOLIDATED FREIGHT CLASSIFICATION RULES NUMBER 21.

(Application for copies should be addressed to Consolidated Freight Classification Committee, 1 Park Avenue, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Materials. The materials used for Cable, Special Purpose, Electrical WM-46/U shall be as specified herein.

3.2 Conductors. Cable, Special Purpose, Electrical WM-46/U shall consist of eight No. 20 AWG conductors and two No. 16 AWG conductors conforming to Specification MIL-W-3861 for coated type B, class K wire for ordinary flexing service.

3.3 Conductor separator. A close wind of fine cotton or rayon shall be applied over each conductor. This separator may be omitted if a free stripping insulating compound is used.

3.4 Insulation.

3.4.1 *Physical properties.* Each conductor shall be insulated with differently colored insulating compounds or the same colored compound depending on the method of color coding (see 3.5). The physical properties of the insulation from each conductor shall conform individually to the requirements for either type IS-L or type IL compound as specified in Specification MIL-I-3930 except that brittleness temperature, torsional stiffness ratio, and cold-tension recovery requirements shall not apply.

3.4.2 *Insulation thickness.* The insulation thickness shall be not less than 0.020 inch, or if followed by a braid or serving over the insulation, the thickness, shall not be less than 0.015 inch.

3.5 Insulation color coding. *Each insulated* conductor shall be differently color coded with colored insulating compounds or a color coded braid. The following colors shall be used on the conductors tabulated below:

Conductor AWG	*Color*
20	Black.
20	White.
20	Red.
20	Green.

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Conductor AWG	*Color (cont'd)*
16	Orange.
16	Blue.
20	Brown.
20	Yellow.
20	Slate
20	Purple

3.6 Shielded conductors. Five of the insulated conductors, colored blue, brown, yellow, slate, and purple, shall each have an individual braid of tinned soft or annealed copper wire applied over the insulation or textile braid if used to a minimum coverage of 85 percent. Each strand of the braid shall consist of No. 34 AWG wire in accordance with Specification MIL–W–3861 for coated type S, soft or drawn-and-annealed wire prior to braiding.

3.7 Cabling. The black and the white conductors shall be twisted together with a lay not exceeding 2 inches. The remaining conductors shall be cabled around the central conductors with a lay not exceeding 7 inches. The outer conductors shall be placed in the same order around the cable as the order of the colors of insulation tabulated in 3.5.

3.8 Overall shield. A close braid of tinned, soft, or annealed copper wire shall be applied over the cable conductors to a minimum coverage of 85 percent. Each strand of the braid shall consist of No. 34 AWG wire in accordance with Specification MIL–W–3861 for coated type S, soft or drawn-and-annealed wire prior to braiding.

3.9 Jacket separator. A close wind of fine cotton or rayon shall be applied over the shield.

3.10 Jacket. The jacket shall be type JS–L in accordance with Specification MIL–I–3930, except that tear resistance, brittleness temperature, torsional stiffness ratio, and cold tension recovery requirement shall not apply. The jacket shall be applied over the separator to an outside diameter of 0.615 plus or minus 0.015 inch. The wall thickness of the jacket shall be not less than 0.030 inch. The outer jacket shall be uniformly smooth and shall have no ridges, pits, depressions, embossed or engraved markings, or any other irregularities which would impair the smoothness of the jacket.

3.11 D–C resistance. When tested as specified in 4.14, each conductor shall have a D–C resistance corresponding to its size at, or corrected to 25° centigrade (C.), as follows:

Conductor size AWG	*D. C. resistance, maximum ohms per 1,000 feet*
20	12.1
16	4.7

3.12 Dielectric strength. The insulated conductor shall withstand for 1 minute without breakdown an alternating current potential of 1,000 volts root mean square (r.m.s.), when tested as specified in 4.15.

3.13 Insulation resistance. Immediately after the insulated conductors have withstood the dielectric strength test, the insulation resistance shall be measured as specified in 4.16. The insulation resistance of Buna S insulated conductors shall be not less than 500 megohms — 1000 feet when measured at or corrected to 15.6°C. The insulation resistance of latex insulated conductors shall be not less than 7,500 megohms — 1,000 feet when measured at or corrected to 15.6°C.

3.14 Cold bend. The insulation and the jacket shall not crack when tested as specified in 4.17 at a temperature of minus 40° plus or minus 2°C.

3.15 Marking of cable. The cable shall be marked on the outside with an inked marking and on the inside under the overall shield with a marker tape. All letters and numbers in the marking shall be of the same height.

3.15.1 *Inked marking.* Inked marking

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shall be clear and legible. The marking shall show no indication of cracking when bent as specified in 3.14. The inked marking shall consist of the designation WM–46/U, and the number of this specification with three to five letter spaces allowed between them. The marking shall be repeated at intervals along the cable not to exceed one foot.

3.15.2 *Marker tape.* The marker tape shall consist of a thin synthetic tape conforming to Specification MIL–I–631 or a flat woven tape of cotton or synthetic yarn. The designation WM–46/U, the number of this specification, the year of manufacture, and the manufacturer's name or trademark, shall be permanently printed in succession on the tape. Serial numbers indicating footage may be supplied at the option of the manufacturer. Spacing between cable designation, number of specification, etc., shall be three letters in width. The marking shall be repeated at intervals along the tape not exceeding one foot. The tape shall be applied longitudinally.

3.16 Workmanship. All cables shall be manufactured and finished in a thoroughly workmanlike manner in accordance with 3.2 through 3.10, and 3.15 through 3.15.2, inclusive.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Acceptance inspection of cable (see 4.4).

(b) Acceptance inspection of preparation for delivery (see 4.21).

4.3 Inspection conditions. Unless otherwise specified herein, all inspection shall be made at room ambient temperature, pressure, and humidity.

4.4 Acceptance inspection of cable. Acceptance inspection shall consist of groups A and B inspection.

4.4.1 *Inspection lot.* An inspection lot shall be as specified in Standard MIL–STD–105.

4.4.2 *Resubmitted lots.* If an inspection lot is rejected, the manufacturer may rework the lot or screen out defectives and resubmit it for acceptance inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lot shall be inspected using tightened inspection.

4.4.3 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table I. Statistical sampling and inspection shall be in accordance with Standard MIL–STD–105.

4.4.4 *Group B inspection.* Group B inspection shall consist of the examinations and tests specified in table II. One specimen shall be taken from each sample unit selected in accordance with the sampling plan specified in 4.4.4.1, except for insulating and jacketing compounds (see 4.19), and conductors and shield strands (see 4.20).

4.4.4.1 *Sampling procedure.* The sampling procedure shall be in accordance with the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL shal be 6.5 percent defective and the inspection level shall be L–7 for normal and tightened inspection and L–5 for reduced inspection.

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TABLE *I. Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL Major	AQL Minor
Visual and mechanical				
Color coding	3.5	4.8	1% for the group	4.0% for the group
Overall shielding	3.8	4.11		
Conductor shield	3.6	4.9		
Cabling	3.7	4.10		
Conductor separator	3.3	4.6		
Jacket separator	3.9	4.12		
Electrical				
D. C. resistance	3.11	4.14	1% for the group	*
Dielectric strength	3.12	4.15		
Insulation resistance	3.13	4.16		

* All electrical defects are considered major.

TABLE II. *Group B inspection.*

Inspection	Requirement paragraph	Inspection paragraph	Sampling plan
Conductor	3.2	4.5	B-2 for group (see 4.4.4).
Insulation thickness	3.4.2	4.7	
Jacket	3.10	4.13	
Cold bend	3.14	4.17	
Marking of cable	3.15	4.18	

4.5 Conductors. Samples of the cable shall be examined for the number and size of conductors as specified in 3.2.

4.6 Conductor separator. Samples of each of the insulated conductors shall be examined for compliance with 3.3.

4.7 Insulation thickness. Insulation thickness shall be measured by any method in Specification J-C-98 for compliance with 3.4.2.

4.8 Insulation color coding. Samples of the cable shall be examined for compliance with 3.5.

4.9 Shielded conductors. Samples of the insulated and shielded conductors shall be examined for compliance with 3.6.

4.10 Cabling. Sampling of the cabled conductors shall be examined for compliance with 3.7.

4.11 Overall shield. Samples of the shielded cable shall be examined for compliance with 3.8.

4.12 Jacket separator. Samples of the cable shall be examined for compliance with 3.9.

4.13 Jacket. Samples of the cable shall be examined for compliance with 3.10.

4.14 Direct current resistance. This test shall be performed in accordance with Method 6021 of Specification J-C-98 for compliance with 3.11.

4.15 Dielectric strength. This test shall be performed in accordance with Method 6111 of Specification J-C-98 for compliance with 3.12.

4.15.1 *Alternate test procedure.* The following alternate test procedure may be used in place of the test procedure specified above:

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(a) Arrange the 10 conductors and the shielding in a rectangle containing 3 rows and 4 columns. The last column will contain 2 unused spaces.

(b) Connect all the conductors in a column together. Do this for each column.

(c) Apply the specified test voltage, for the specified time, between each column of the rectangle in turn, and the remaining columns connected together.

(d) Connect all the conductors in a row together. Do this for each row.

(e) Apply the specified test voltage, for the specified time, between each row of the rectangle in turn, and the remaining rows connected together.

(f) Connect all the conductors together.

(g) Apply the specified voltage, for the specified time, between all the conductors connected together and to sheath.

No. of conductors	*No. rows*	*No. columns*	*No. unused spaces*
10	3	4	2

4.16 Insulation resistance. This test shall be performed in accordance with Method 6031 of Specification J–C–98 for compliance with 3.13.

4.17 Cold bend.

4.17.1 *Specimens.* One specimen from each sample unit shall be prepared for test. In the event of failure of any one specimen, two additional specimens shall be prepared.

4.17.1.1 *Procedure.* The specimens selected for checking cable as a whole shall be attached to a 2.38 inch mandrel. The specimens removed from the cable for checking the insulation apart from the cable shall consist of the insulated conductors with all other coverings over the insulation removed. One of each differently color insulated conductor shall be attached to the proper size mandrel. The No. 16 AWG insulated conductor shall be attached to a .094-inch mandrel and the No. 20 AWG insulated conductor shall be attached to a .062-inch mandrel. The specimens shall be suspended vertically with lower ends weighted sufficiently to keep specimens taut and to prevent bending them without handling. The mandrel and specimens shall be placed for at least 20 hours in the cold chamber at a temperature of minus 40 plus or minus 2°. While at this temperature, the specimens shall be bent for five close turns around the mandrel at the rate of approximately 15 turns per minute. After the test has been completed, the jacket on the specimen of cable shall be examined through a magnifying glass of at least three-diameter magnification and then removed. The conductor insulation on all specimens shall be examined for cracks with the magnifying glass for compliance with 3.14.

4.18 Marking of cable. Samples of the cable shall be examined for compliance with 3.15.

4.19 Insulating and jacketing compounds. The insulating and jacketing compounds shall be examined and tested as specified in Specification MIL–I–3930.

4.20 Conductors and shield strands. The conductors and shield strands prior to fabrication of cable shall be examined and tested as specified in Specification MIL–W–3861.

4.21 Acceptance inspection of preparation for delivery.

4.21.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL–P–116. Classification of defeats shall conform to tables III and IV.

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TABLE III. *Classification of defects for preservation, packaging and marking.*

Major	Minor
1. Quantity per unit package not in accordance with the contract.	1. Item not properly blocked or braced within the unit package to prevent movement.
2. Incorrect packaging method applied	2. Packaging material damaged.
3. Stock number omitted, incorrect or illegible ..	3. Any item of marking information other than 3 through 5 listed under major defects missing or illegible.
4. Nomenclature incorrect or missing	
5. Quantity of items in package not indicated ...	

TABLE IV. *Classification for packing and marking.*

Major	Minor
1. Box closure not in accordance with specification.	1. Unsealed carton.
2. Type, grade, class and style of the shipping container do not meet the minimum requirements.	2. Defective taping or sealing of carton.
3. Strapping when required is omitted or inadequate, or incorrectly applied.	3. Any other box defect which may be considered minor by definition of MIL–STD–105.
4. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement and/or damage.	4. Any item of required marking information, other than 8 through 14 listed under major defects, missing or illegible.
5. Gross weight in excess of contract limitation..	
6. Use of improper or defective material	
7. Shipping documents or packing list omitted (when required).	
8. Stock number (when required) omitted, incorrect, or illegible.	
9. Nomenclature (when required) omitted or illegible.	
10. Quantity incorrect or missing	
11. Destination marking incorrect or illegible ...	
12. Service designation (color marking) missing.	
13. Special marking and labeling requirements cited in the contract, such as MAP labels, shipment digit markings, etc. not applied.	
14. Overseas code marking—illegible, incomplete or missing.	

4.21.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed in tables III and IV. This inspection shall conform to the appendix to Standard MIL–STD–105. Inspection level L–S shall be used for normal inspection and L–6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL for major

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defects shall be four percent and the AQL for minor defects shall be ten percent.

4.21.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.21.2.2 *Procedure in case of failure.* If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken.

4.21.2.3 *Disposition of nonconforming product.* Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL–STD–105.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Cable Special Purpose, Electrical WM–46/U shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and described herein.

5.1.1.1 *Cable, Special Purpose, Electrical WM–46/U.* Special purpose cable in the length specified in the bid request or contract shall be packaged as follows.

5.1.1.1.1 *Short lengths.* Special purpose cable not of sufficient length to be placed on reels shall be packaged Method IC–5, as follows: Each length of cable shall be wound into a coil of the smallest proportionate dimensions. The diameter of the inside eye of the coil shall be large enough to prevent damage to the cable from winding. Tie the coil in four places with cotton tape, conforming to Specification DDD–T–86, securing the ends to the coil with the ties. Place each coiled cable within a close-fitting, RSC, type I, class 2, grade 5 box conforming to Specification PPP–B–636. Closure shall be in accordance with the appendix of the box specification using tape conforming to Specification PPP–T–60, type III, class 1, olive drab.

5.1.1.1.2 *Long lengths.* Special purpose cable of sufficient lengths to be reeled shall be packaged in accordance with Specification MIL–R–10409 or Specification MIL–R–12083.

5.1.2 *Level C.* Cable. Special Purpose, Electrical WM–46/U shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* A quantity of Cable, Special Purpose, Electrical WM–46/U, packaged as specified in 5.1.1.1.1 shall be packed within overseas class or type wood wirebound boxes, fiberboard, wood-cleated boxes, wood, cleated plywood boxes, fiber boxes, or nailed wood boxes, conforming to Specifications PPP–B–585, PPP–B–591, PPP–B–601, PPP–B–636 or PPP–B–621, as applicable. The weight of any box of contents shall not exceed 200 pounds except fiber boxes when the weight of the box with contents shall not exceed 70 pounds. Box closure shall be in accordance with the applicable box specification and appendix thereto.

5.2.1.1 *Reels.* Reels shall be packed in accordance with Specification MIL–R–10409 or Specification MIL–R–12083.

5.2.2 *Level B.* Cable, Special Purpose, Electrical WM-46/U shall be packed for shipment within domestic class or type, fiberboard, wood cleated boxes, wood, cleated plywood boxes, nailed wood boxes, or corrugated triple wall boxes conforming to Specifications PPP–B–591, PPP–B–601,

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PPP–B–621, or PPP–B–640, as applicable. The weight of any one box with contents shall not exceed 200 pounds. The box closure shall be in accordance with the applicable box specification and appendix thereto.

5.2.2.1 *Reels.* Reels shall be packed as specified in 5.2.1.1.

5.2.3 *Level C.* Cables packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification rules, Motor Freight or Parcel Post regulations whichever is applicable.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL–STD–129.

6. NOTES

6.1 Intended use. Cable, Special Purpose, Electrical WM–46/U is intended for use as a power or audio frequency interconnecting cable between communication equipments at potentials up to 300 volts root-mean-square.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (level A, level B, or level C).

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

6.3 Definitions. For the purpose of this specification, the definitions given below apply.

6.3.1 *Inspection lot.* An inspection lot consists of cable produced under substantially the same conditions and offered for inspection at one time.

6.3.2 *Lot size.* The lot size is the number of units of product in an inspection lot.

6.3.3 *Unit of product.* The unit of product is a continuous length of finished cable contained on a reel or spool, or in a coil.

6.3.4 *Specimen.* A specimen is a single piece of finished cable or any part removed from the finished cable such as conductors, separators, insulated conductors, shielded conductors, or cabled and shielded conductors, insulations, and jacket, which is taken from a sample unit and subjected to inspection.

6.3.5 *Finished cable.* Finished cable is a cable on which all manufacturing operations have been completed and which is ready to be submitted for acceptance.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder, or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
AF—Air Force

Other interest:
Army—QO

Preparing activity:
Army—Signal Corps

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